

SERUM: State Extraction and Refinement for User Modeling

Andy J. Phu, James Mooney, Karin de Langis, Khanh Chi Le, Dongyeop Kang

Minnesota NLP Lab

University of Minnesota

Minneapolis, MN 55455, USA

{phu00004, moone174, dento019, le000422, dongyeop}@umn.edu

Abstract

Agentic assistants capable of proactive, personalized interactions require structured models of user intent and workflow. However, building these models from raw, unstructured screen activity remains an open challenge. We present SERUM, a multi-pass framework that extracts finite-state behavioral models directly from unstructured egocentric video using hierarchical VLM annotation. Processing screen recordings through a sliding window, SERUM alternates between activity-recognition and intent-inference passes, with each pass refining labels using accumulated prior context to reduce hallucination and temporal conflation seen in single-pass annotation. Synonymous states are then merged via sentence embeddings and human-calibrated thresholds into a compact, coherent taxonomy. We evaluate behavioral structure by fitting first-order Markov models over the resulting label sequences (both actions and intents) and measuring predictive accuracy against frequency baselines. Across 61 egocentric videos in four domains (coding, cooking, physical activities, and daily life), we find: (1) iterative label refinement converges to a stable state vocabulary, which we term *schematic equilibrium*, after several passes; (2) normalized Markov models achieve substantially lower perplexity and higher action predictions than frequency baselines, with the largest gains on structured tasks like coding; and (3) human annotators rate final-pass labels as accurate and meaningfully improved over first-pass labels. To our knowledge, SERUM is the first system to produce interpretable process models from unstructured egocentric screen video without manual annotation, opening a scalable pathway for user modeling and behavioral understanding in the wild. Our demo, code, and results are publically available ¹

1 Introduction

Proactive AI assistants need structured models of user behavior — compact representations of how people move through goal-directed activity over time. Before platforms like YouTube, and Twitch, rich egocentric footage has been scarce. Furthermore, converting raw video into structured behavioral models is non-trivial. Existing activity recognition methods either depend on fixed hand-crafted taxonomies (Damen et al., 2022; Grauman et al., 2022) or require expensive frame-level annotation. Process mining produces elegant behavioral models from event logs (van der Aalst et al., 2012; van der Aalst, 2016), but assumes activity labels already exist. Neither path applies to unstructured, open-ended video.

We ask: *can we extract interpretable, structured models of user behavior directly from raw egocentric video, without a predefined ontology and without manual annotation?*

A simple answer is to prompt a vision-language model (VLM) to label each frame, similar to recent work on general user models Shaikh et al. (2025). But single-pass annotation fails in two ways: VLMs hallucinate, and they suffer from *temporal conflation* — collapsing semantically distinct activities into generic labels because they lack surrounding context.

¹<https://minnesotanlp.github.io/SERUM-web/>

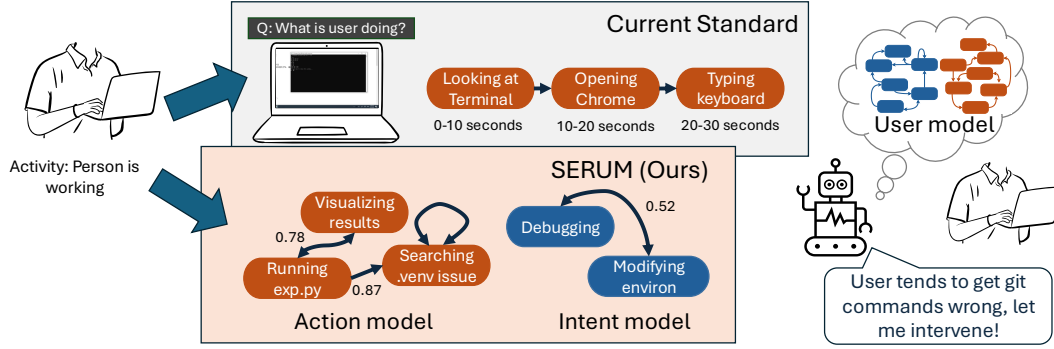


Figure 1: (Top) Current standard methods process each frame independently, producing isolated activity descriptions and coarse intent estimates. (Bottom) SERUM’s multi-pass pipeline revisits prior context across frames, enabling the construction of a refined user model for both user actions and intents, enabling better informed proactive suggestions.

We introduce **SERUM** (State Extraction and Refinement for User Modeling), a multi-pass pipeline that addresses these limitations through alternating rounds of *activity recognition* and *intent inference*, each grounded in the accumulated context of prior passes. SERUM operates at two complementary levels: *user actions* (directly observable behaviors, e.g., pulling a git repository”) and *user intents* (intermediate goals, e.g., setting up a development environment”). A sliding context window provides each pass with a run-length encoding of surrounding frames. After annotation, a *label normalization* step merges synonymous labels into a compact vocabulary. SERUM outputs activity and intent models implemented as first-order Markov chains, capturing the probabilistic transition structure of user behavior.

We apply SERUM to 61 egocentric YouTube videos spanning coding, cooking, physical activity, and daily life. Our experiments show that: (1) the extracted label vocabulary reliably converges to a stable taxonomy by pass 8 — a phenomenon we term *schematic equilibrium*; (2) label normalization compresses the vocabulary and sharpens transition structure, yielding better predictive models; (3) normalized Markov models outperform frequency baselines on both action and intent sequences; and (4) annotators judge final-pass labels as correct 88.3% of the time ($\alpha=0.40$) and prefer them over first-pass labels 82.8% of the time ($\alpha=0.41$).

SERUM is, to our knowledge, the first framework for extracting structured user activity and intent models directly from unstructured egocentric video — requiring no logs, no predefined taxonomies, and no labeled data. The resulting models represent an early validation for downstream applications such as proactive agentic assistance and personalization. Code and data are publicly available.²

2 Related Work

Egocentric Video Understanding and Action Anticipation. Benchmarks such as EPIC-KITCHENS (Damen et al., 2018; 2022) and Ego4D (Grauman et al., 2022) have established that predicting what a user will do next requires reasoning at two levels: the immediate action and the underlying goal. Mascaro et al. (Mascaro et al., 2023) exploit this hierarchy by conditioning low-level action predictions on inferred high-level intentions for long-term anticipation. Furnari & Farinella (2020) show that rolling-unrolling recurrent representations further improve anticipation on EPIC-KITCHENS. SERUM is complementary: rather than operating on labeled benchmark data, it infers both action and intent labels from scratch using raw, unannotated video.

VLMs as Video Annotators and User Modelers. Foundation vision-language models such as the Qwen-VL family (Bai et al., 2023; Wang et al., 2024; Bai et al., 2025) have lowered the cost of video annotation, but single-pass inference on long videos suffers from hallucination

²<https://github.com/minnesotanlp/SERUM>

and temporal conflation. ROVER (Schroeder et al., 2025) and VideoNarrator (Wu et al., 2025) address these failures through recursive decomposition and multi-component verification pipelines, respectively; LLMs more broadly have been shown to match or surpass crowd workers when labels are iteratively verified (Gilardi et al., 2023; He et al., 2024). SERUM shares the insight that iterative, context-aware annotation suppresses hallucination, but goes further: each pass re-annotates frames conditioned on all prior-pass labels, enabling the model to revise earlier judgments as context accumulates and driving convergence toward a self-consistent vocabulary without any predefined ontology. Most closely related in motivation is GUM (Shaikh et al., 2025), which builds user models from computer-use screenshots by inferring and revising confidence-weighted propositions about user preferences and knowledge. While GUM targets *who* the user is, SERUM targets *what* the user is doing and intends to do next, producing activity and intent transition models via multi-pass re-annotation rather than propositions about user traits.

Process Mining and Behavioral Sequence Models. Process mining recovers structured process models from event logs (van der Aalst et al., 2012; van der Aalst, 2016), but these algorithms assume clean logs of known event types. SERUM, while conceptually related, assumes no ontology of event types and uses unstructured video as input. Process-mining quality criteria (i.e., fitness, precision, and generalization (Buijs et al., 2012)) motivate our use of next-action prediction accuracy and perplexity as evaluation metrics.

Semantic Label Normalization. Open-vocabulary annotation produces synonymous labels that inflate state-space size and degrade model quality. We consolidate them via pairwise embedding similarity using Sentence-BERT (Reimers & Gurevych, 2019a), with a human-calibrated merging threshold ($t^* = 0.4268$). This is analogous to entity resolution and ontology alignment in knowledge-base construction. We treat normalization as a design component rather than a post-hoc fix, and empirically show it improves both vocabulary compactness and predictive accuracy of the resulting Markov models.

3 SERUM: State Extraction and Refinement for User Modeling

SERUM is a multi-pass framework for extracting structured behavioral models from raw ego-centric video—no predefined ontology, no manual annotation required. Given a sequence of T sampled frames $\{f_1, \dots, f_T\}$, SERUM alternates between grounded activity recognition and intent inference, progressively refining coarse perceptual observations into temporally coherent behavioral descriptions. The final output is a pair of **User Models** (UMs)—one over actions, one over intents—that compactly represent the user’s behavioral dynamics and directly support downstream applications such as proactive next-action prediction. Figure 2 provides an overview of the full pipeline.

3.1 Actions and Intents

SERUM represents behavior at two distinct levels. **Actions** (a_t) are mid-level natural-language descriptors of directly observable behavior at frame t (e.g., “washing vegetables,” “pulling a git repository”). **Intents** (i_t) are latent goal-directed states, that may not be directly observable, but are inferred from sequences of actions (e.g., “preparing dinner,” “setting up a dev environment”).

The two levels are *mutually informative*: action evidence anchors intent inference; intent context disambiguates ambiguous actions. For example, “looking at a phone” is labeled “checking map directions” once intent context establishes “navigating to a destination.” This bidirectionality motivates SERUM’s alternating design—running separate independent passes for each level underperforms because neither level grounds the other. Prompt templates are provided in Appendix G.

3.2 Multi-Pass Annotation Pipeline

A natural baseline is single-pass VLM annotation: prompt the model once per frame for both labels. This fails for two reasons: intent inference is inherently *retrospective* (the meaning of

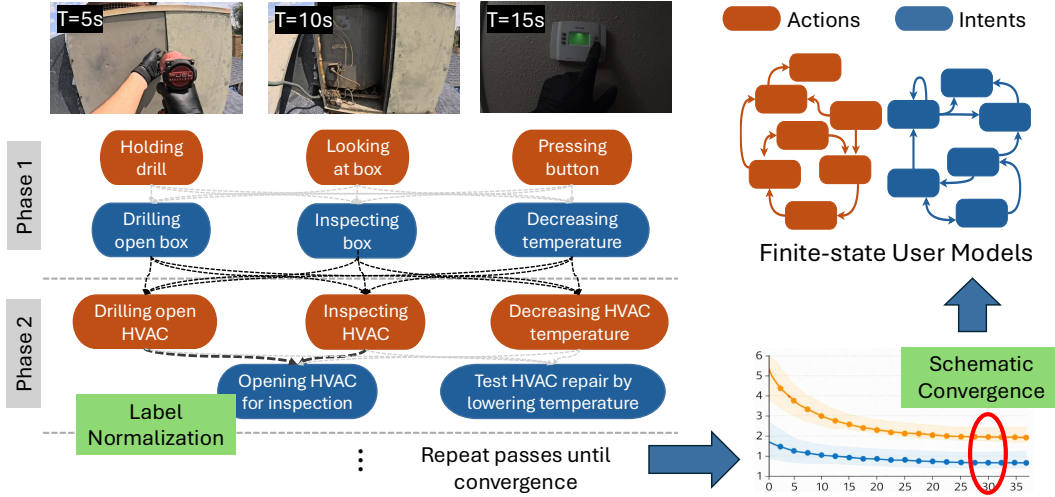


Figure 2: The SERUM pipeline applied to an HVAC repair video. Frames are annotated through alternating activity (orange) and intent (blue) passes. Early passes yield generic labels (e.g., “holding drill”); later passes produce fine-grained, context-aware labels (e.g., “testing HVAC repair”). Labels are normalized before constructing the final User Models.

an action often only becomes clear after observing subsequent frames), and without shared context across frames, VLMs produce *semantically inconsistent* labels (e.g., “rinsing produce” vs. “cleaning vegetables” for the same activity). Running separate independent passes for each level does not help either. Actions and intents are *mutually informative*—intent context disambiguates ambiguous actions, while action evidence anchors intent inference. Only by *alternating* the two—each pass conditioning on the outputs of the last—can they ground each other in a feedback loop converging toward coherent, disambiguated descriptions.

SERUM implements this via alternating passes over frames $\{f_1, \dots, f_T\}$ using Qwen3-VL-8B-Instruct, where odd passes annotate actions and even passes annotate intents. Pass 1 produces unconditioned action labels $\{a_t^{(1)}\}$ in free-form natural language. Pass 2 infers intent labels $\{i_t^{(1)}\}$ conditioned on $\{a_t^{(1)}\}$; Pass 3 refines action labels conditioned on $\{i_t^{(1)}\}$; and so on. Each pass (from Pass 2 onward) receives two context signals. First, a **temporal context window** of $w=20$ neighboring frames, encoded as a run-length encoding (RLE) that collapses consecutive identical states into count-weighted entries, providing dense local context without exceeding the model’s token limit. Second, an **inter-pass summary**—a natural-language summary generated from the full-pass RLE and the prior summary—that propagates global narrative context forward as compressed episodic memory. For long videos exceeding 2^{15} tokens, a map-reduce procedure summarizes fragments independently before merging into a coherent global summary. Passes continue until the label vocabulary stabilizes—empirically by pass 8—a convergence we term *schematic equilibrium* (§4.2).

3.3 Label Normalization

Free-form annotation produces surface synonyms that inflate vocabulary size and degrade model quality. We resolve these via Sentence-BERT (Reimers & Gurevych, 2019a) cosine similarity with a human-calibrated merging threshold $t^*=0.4268$, chosen to maximize F1 on human-judged synonym pairs.³ This reduces vocabulary size by 46.0% on average and measurably improves predictive accuracy downstream.

³Calibration procedure detailed in Appendix F.

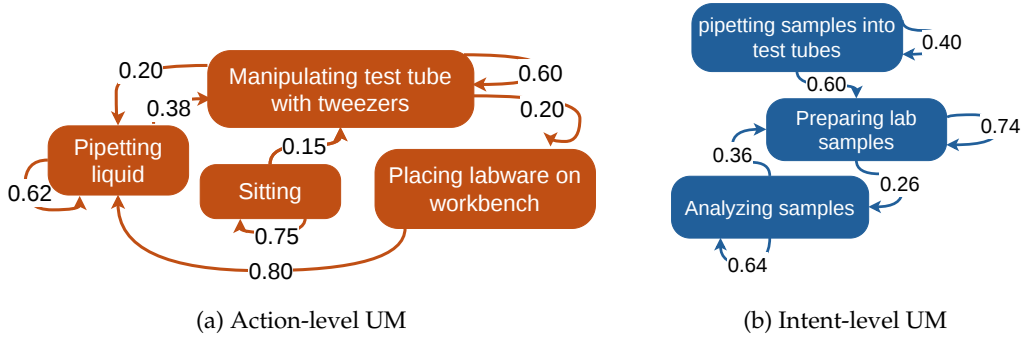


Figure 3: Example User Models from a single video. Action-level states (a) capture observable behaviors; intent-level states (b) capture inferred goals.

Table 1: Per-domain dataset statistics. Vocabulary and accuracy from the final activity (P11) and intent (P12) passes. Values are mean $\pm$ std. Act. = Activity, Int. = Intent. We find that software engineering videos tend to consist of repetitive actions (writing code) focused on a singular goal (releasing a snake game), resulting in higher accuracy.

Domain	Videos	Frames	Act. Vocab	Int. Vocab	Act. Acc (%)	Int. Acc (%)
Coding	19	207 $\pm$ 85	7 $\pm$ 3	26 $\pm$ 22	73.9	47.5
Cooking	15	219 $\pm$ 101	66 $\pm$ 29	48 $\pm$ 24	13.6	21.8
Physical	12	155 $\pm$ 81	29 $\pm$ 17	24 $\pm$ 11	25.7	40.1
Daily Life	15	136 $\pm$ 81	28 $\pm$ 21	39 $\pm$ 26	24.7	21.5
All	61	182 $\pm$ 94	31 $\pm$ 29	34 $\pm$ 24	37.5	33.3

61 videos, 11,125 frames, 927 min (15.5 hrs), 12 passes each, 133,500 total state extractions. A generalization study on EPIC-KITCHENS-100 (366 videos, 37 participants) is reported in Appendix C.

3.4 Output: User Models

The final output is a pair of **User Models** (UMs): directed weighted graphs $\mathcal{M}=(\mathcal{S}, \mathcal{E})$ where $\mathcal{S}$ is the canonicalized state vocabulary and each edge (s, s') is weighted by observed transition frequency. One UM is built over actions, one over intents, yielding complementary views of behavior (Figure 3). UMs support proactive assistance by surfacing probable next states given the user’s current state. We evaluate predictive utility via a **next-action prediction** task: UMs trained on the first 60% of frames predict the held-out final 40%, measuring whether captured behavioral dynamics generalize to unseen activity.

4 Evaluation

We evaluate SERUM on 61 egocentric videos spanning coding, cooking, physical activity, and daily life, addressing four research questions: **RQ1** (§4.2) does SERUM’s iterative annotation converge to a stable label vocabulary (*schematic equilibrium*)? **RQ2** (§4.3) do the resulting user models usefully predict next user states? **RQ3** (§4.4) are SERUM’s labels aligned with human judgment? **RQ4** (§4.5) how do label normalization and intent passes each contribute to model quality?

4.1 Experimental Setup

Dataset We evaluate on 61 egocentric videos sampled at 5-second intervals, yielding 11,125 total frames across four domains (Table 1).

Model and inference. All annotation passes use Qwen3-VL-8B-Instruct (Bai et al., 2025)⁴ in BF16, served via vLLM (Kwon et al., 2023) with tensor parallelism across two GPUs per node. We distribute inference across two nodes (2× NVIDIA A5000, 2× NVIDIA A6000), each running an independent vLLM server, achieving a combined throughput of 1.3 inferences/sec and processing 12 passes for a 10-minute video in ≈ 17 minutes per node.

Label Normalization. We apply pairwise semantic merging using SentenceBERT embeddings (Reimers & Gurevych, 2019b) with cosine-distance threshold $t^* = 0.43$, selected to maximize F1 on a human-annotated calibration set of 100 activities and 100 intents randomly sampled from all passes. (§6a).

4.2 RQ1: Schematic Equilibrium

Setup. To determine how many annotation passes are needed, we ran a pilot study on 13 videos for 30 passes, tracking vocabulary size per pass before and after label normalization.

Results. As shown in Figure 4, the average raw activity vocabulary drops from ~ 28 to ~ 18 unique states by pass 8, while the average raw intent vocabulary drops more steeply from ~ 54 to ~ 24 . Label normalization compresses both further to ~ 10 states and remains stable thereafter. Per-video vocabulary curves (Figure 4c) confirm that all videos individually stabilize by pass 8 — a convergence we term *schematic equilibrium*. Based on this finding, we run all large-scale evaluations at 12 passes (6 activity and 6 intent, interleaved), providing a margin beyond the observed convergence point.

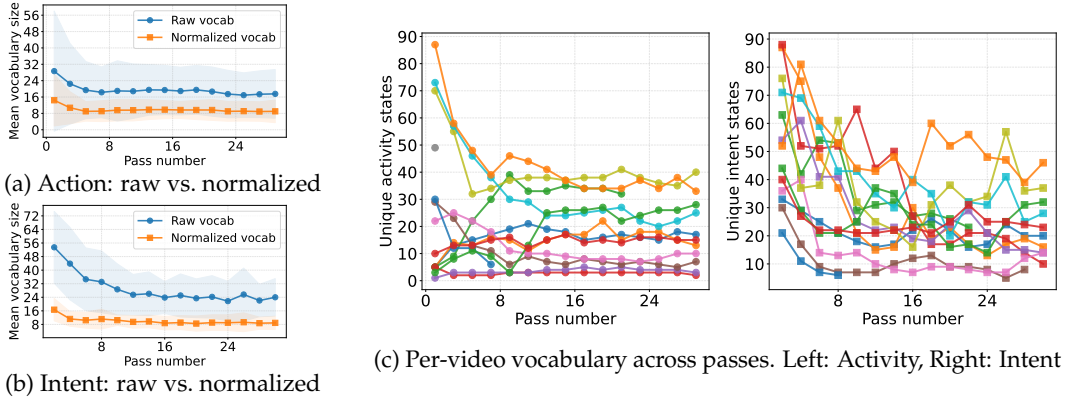


Figure 4: 30-pass pilot study (13 videos). (a) Activity vocabulary drops from 28 to 18 raw states by pass 8, with normalization compressing further to 10. (b) Intent vocabulary shows a steeper decline from 54 to 24 raw states, with normalization consistently reducing to 10 across all passes. (c) Per-video vocabulary stabilizes by pass 8 (*schematic equilibrium*).

4.3 RQ2: Next-State Prediction

Setup. We construct Markov user models from the first 60% of frames per video and evaluate on the remaining 40%, applying add-one Laplace smoothing to transition counts. We compare against three baselines: **Majority** (always predict the most frequent training state), **Weighted Random** (sample proportional to marginal frequency), and **Uniform** (sample uniformly over observed states). Performance is measured by **top-1 accuracy** and **perplexity** (exponentiated cross-entropy over held-out transitions; lower is better).

Main Result. Table 2 shows that at the final annotation pass, Markov user models outperform naive baselines in both top-1 accuracy and perplexity.⁵ Normalized variants (superscript n) apply post-hoc label normalization merging prior to model construction.

⁴Preliminary study on model choice in Appendix H

⁵Top-3 and Top-5 show similar tendencies but more strongly favor Markov.

Table 2: Mean top-1 accuracy and perplexity at the final pass. ⁿ denotes normalized labels.

Model	Activity		Intent	
	Top-1 ($\uparrow$)	PPL ($\downarrow$)	Top-1 ($\uparrow$)	PPL ($\downarrow$)
Markov	37.5 $\pm$ 33.3	23.9 $\pm$ 25.6	33.3 $\pm$ 26.9	24.6 $\pm$ 21.5
Majority	37.6 $\pm$ 34.0	29.8 $\pm$ 32.0	29.9 $\pm$ 28.1	31.8 $\pm$ 35.5
Wt. Random	25.9 $\pm$ 29.5	29.8 $\pm$ 32.0	17.9 $\pm$ 20.4	31.8 $\pm$ 35.5
Uniform	9.2 $\pm$ 10.5	30.8 $\pm$ 29.0	6.5 $\pm$ 9.0	34.2 $\pm$ 24.3
Markov ⁿ	48.5$\pm$30.9	10.9$\pm$11.0	58.2$\pm$30.3	8.2$\pm$ 9.7
Majority ⁿ	46.5 $\pm$ 32.6	14.4 $\pm$ 15.8	53.4 $\pm$ 33.8	11.0 $\pm$ 14.7

Normalization benefits the Markov model disproportionately, since fewer labels reduces sparsity in its transition matrix, whereas the Majority baseline only tracks label frequencies.

Table 3: Markov top-1 accuracy and perplexity by domain (final pass, normalized labels).

Domain	n	Act Top-1 (%)		Intent Top-1 (%)		Perplexity ($\downarrow$)	
		Raw	Norm.	Raw	Norm.	Activity	Intent
Coding	19	73.9	76.4	47.5	76.1	3.1	17.4
Cooking	15	13.6	31.8	21.8	53.8	53.2	34.8
Physical	12	25.7	44.2	40.1	71.6	22.2	14.9
Daily Life	15	24.7	33.5	21.5	30.3	22.5	31.3
Overall	61	37.5	48.5	33.3	58.2	23.9	24.6

Domain-level results (Table 3) show largest gains on structured tasks: Coding achieves 76.4% normalized activity accuracy, reflecting the rich, repetitive transition structure of coding workflows. Cooking and physical tasks see smaller absolute accuracy but substantial relative gains from normalization. We further find preliminary evidence that SERUM-produced markov models can transfer to similar but unseen workflow videos. ⁶

4.4 RQ3: Human Assessment of Label Quality

Setup. We recruited five colleagues with domain expertise in human workflow research to evaluate label quality. Annotators assessed 180 uniformly sampled frames from nine videos (10 activity, 10 intent per video) presented via a web application embedding the source video at the relevant timestamp. For each frame, annotators judged: (1) **label accuracy** — whether the final-pass label correctly describes the observed activity or intent; and (2) **pass preference** — whether the final-pass or first-pass label is better. Responses are aggregated by majority vote; inter-annotator agreement (IAA) is measured via Krippendorff’s α .

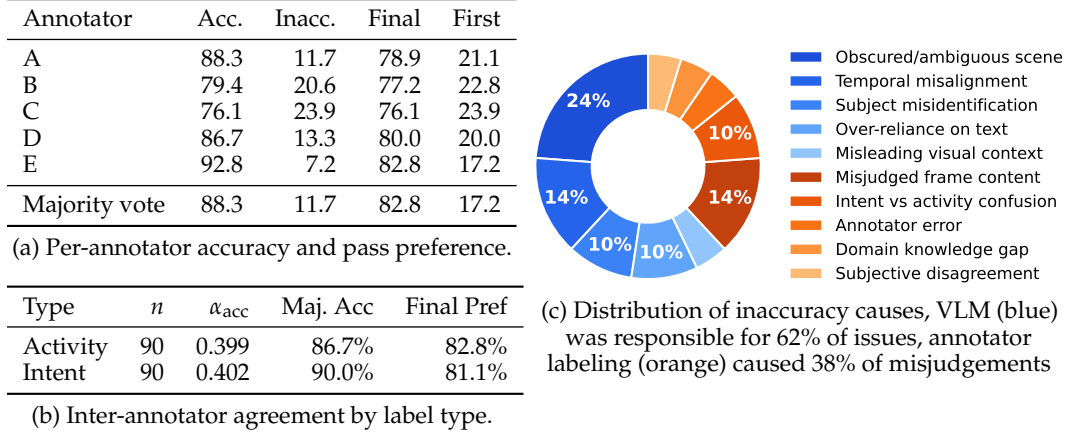
Results. By majority vote, **88.3%** of labels were rated accurate ($\alpha = 0.40$) and final-pass labels were preferred over first-pass labels **82.8%** of the time ($\alpha = 0.41$), suggesting that iterative annotation produces meaningfully better labels (Figure 5). Intent labels were slightly more accurate than Activity labels (90% vs. 86.7%), yet preference rates were comparable across both types (81.1% vs. 82.8%), indicating that multi-pass refinement improves intent inference similarly to activity recognition despite intent being harder to verify from a single frame.

4.5 RQ4: Ablation Study

Setup. We isolate the contribution of the three core design choices in SERUM. For **label norm**, we compare Markov models built on raw vs. normalized label sequences at the final pass, measuring vocabulary reduction, top-1 accuracy, and perplexity. For **intent passes**, We compare the full SERUM pipeline against an activity-only baseline pipeline in three scenarios. (1) both pipelines infer and evaluate on their own freely generated vocabularies,

⁶Preliminary study detailed in Appendix I.

Figure 5: Human annotation results and error analysis



(2) both pipelines initially infer on open vocabularies, but project activity-only labels on to the intent-conditioned vocabulary before evaluation, (3) Project in reverse direction. For **temporal window size**, we compare SERUM’s default temporal window of $n=20$ with $n=10$ and $n=0$ on accuracy and perplexity against respective baselines.

Label normalization. Normalization reduces the state vocabulary by **46.0%** on average ($\pm 21.1\%$) while improving Markov top-1 accuracy by **+18.2 pp** (± 21.2 pp) and reducing perplexity by 14.9 points. Gains are consistent across domains and largest for intent models, where surface-synonym proliferation is most severe. This validates open-vocabulary annotation followed by principled merging as better than a fixed ontology: the former preserves fine-grained behavioral distinctions that the latter would collapse, and normalization then recovers the compact transition structure needed for reliable Markov estimation. The calibrated SentenceBERT threshold ($t^* = 0.43$, $F1 = 0.822$; Figure 6b and 6c) separates synonymous from distinct labels with high precision (0.768) and recall (0.883).

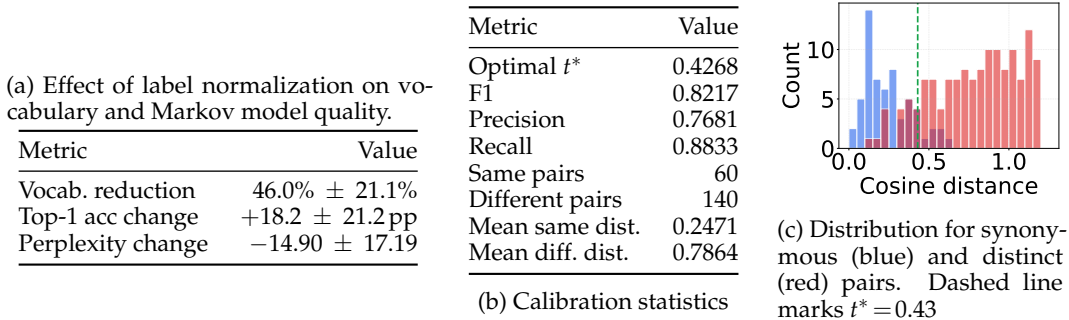


Figure 6: Label normalization summary (left) and semantic threshold calibration on 200 human-annotated label pairs (right).

Value of intent passes. To isolate the contribution of intent gathering annotation passes, we compare the full pipeline against an activity-only baseline using 12 activity passes with no intent inference. The preference study below tests whether this predictability reflects genuine label quality. Across 61 videos, 47% of frames received different activity labels between the two conditions. We sampled 30 of these divergent frames (10 per domain, stratified across 3 videos) and presented each as a blinded A/B pair.

By majority vote, annotators preferred labels from the full pipeline **73%** of the time ($\alpha = 0.726$; Table 4). Without intent context, activity labels collapse to uninformative dominant states: `typing_on_keyboard` for 92–96% of coding domain frames (vs. the intent-informed pipeline’s `editing_css_style`, `editing_html_code`, `debugging_code`).

Table 4: Ablation preference study: full pipeline (activity+intent) vs. activity-only labels. Three annotators evaluated 30 blinded A/B pairs across three domains.

Annotator	n	Full Pref. (%)	Act.-Only Pref. (%)	Metric	Value
A	30	80	20	Krippendorff's α	0.726
B	30	73	27	A vs B agree	93%
C	30	63	37	A vs C agree	83%
Majority	30	73	27	B vs C agree	90%

Value of intent passes with frozen vocabularies. To study the contribution of intent passes, notwithstanding differences in vocabulary size produced by the activity-only and the full pipelines, we project the vocabulary produced by the full pipeline onto the activity-only pipeline vocabulary, project the activity-only pipeline onto the full pipeline vocabulary, and test the respective performances of both vocabularies. Both normalizations support the conclusion that vocabulary size differences do not significantly impact results.

Table 5: full vs. activity-only vocabulary before and after projection. OOV measures percent of labels that have no match in target vocabulary.

Condition	Vocab	Markov	Majority	PPL	OOV
<i>Raw (own vocabulary)</i>					
Intent-conditioned	29.9±27.4	36.4±30.8	36.8±32.1	22.5±23.1	—
Activity-only	28.0±26.8	42.6±34.0	43.6±34.2	20.7±21.7	—
<i>Normalized $\rightarrow$ intent vocab</i>					
Intent-conditioned	18.4±14.8	45.9±28.9	42.6±30.4	11.9±10.9	1.2
Activity-only	17.2±14.5	50.2±31.0	46.5±33.5	11.1±10.8	2.0
<i>Normalized $\rightarrow$ activity-only vocab</i>					
Intent-conditioned	16.9±13.6	48.4±28.0	44.6±29.9	10.6±10.0	7.2
Activity-only	17.8±15.2	50.0±31.2	46.8±33.3	11.3±11.5	1.2

Effect of temporal window size. To study the effects of various temporal window sizes, we evaluate over the same 12 randomly chosen videos at window sizes $n = \{0, 10, 20\}$. Markov - Majority gap increases at higher window size (4.4 vs 8.0), suggesting prediction structures become more prominent in produced Markov models at higher window sizes.

Table 6: Temporal-window sensitivity ($w \in \{0, 10, 20\}$), $n = 12$ videos. Markov / Majority in %. Normalized: all conditions projected onto the $w = 20$ vocabulary.

Condition	Avg. vocab	Markov	Majority	Perplexity
<i>Raw vocab</i>				
$w = 0$ (no temporal context)	63.00 ± 32.74	19.2 ± 25.3	19.3 ± 25.5	47.76 ± 24.55
$w = 10$	55.08 ± 29.10	19.8 ± 23.9	16.6 ± 25.9	41.51 ± 20.32
$w = 20$ (paper default)	54.50 ± 26.94	21.5 ± 23.8	19.4 ± 25.3	40.53 ± 19.64
<i>Normalized</i>				
$w = 0$	30.08 ± 14.63	28.4 ± 24.3	24.0 ± 25.8	20.38 ± 11.86
$w = 10$	27.92 ± 13.69	28.0 ± 24.9	21.7 ± 27.0	18.98 ± 11.61
$w = 20$	30.75 ± 14.68	32.6 ± 27.3	24.6 ± 25.5	20.46 ± 11.97

Qualitative Study. Figure 7 shows multi-pass refinement and its predictive consequence on a coding video.⁷

⁷Additional examples in Appendix, Figure 8.

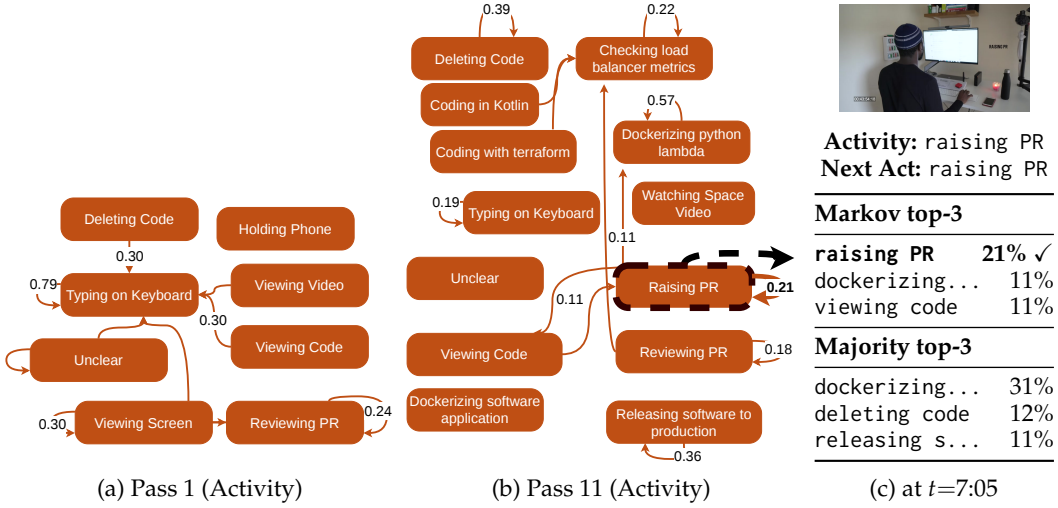


Figure 7: Activity refinement and next-state prediction for behindP12. (a) Pass 1 produces generic labels. (b) By pass 11, task-specific states emerge. (c) The Markov model correctly predicts state persistence by conditioning on the current state, while the majority baseline erroneously predicts the three globally most frequent states regardless of context.

5 Conclusion and Discussion

We presented SERUM, a multi-pass VLM framework that extracts structured activity and intent models from egocentric video without a predefined ontology or manual annotation. Alternating activity and intent passes converge to a stable vocabulary (*schematic equilibrium*) by pass 8; subsequent label normalization compresses it by 46%, yielding Markov user models that outperform frequency baselines on next-state prediction. Human annotators rate 88.3% of final-pass labels accurate and prefer them over first-pass labels 82.8% of the time, suggesting that iterative refinement produces meaningful, recognizable improvements. This work has the following limitations and interesting directions for future work:

Evaluation protocol. Split-half evaluation penalizes Markov models on videos whose content progresses linearly without revisiting earlier states, since training and test vocabularies become largely disjoint. This effect can be seen with several videos achieving near-zero accuracy before normalization (§8). Future work could address this through cross-video evaluation, where models trained on one user’s videos predict states in another’s.

Downstream applications. An important open question is whether SERUM’s user models can drive proactive agentic assistance — anticipating recurring errors or context switches before they occur. Although SERUM’s computational complexity presents a challenge in latency to its feasibility in live settings, the majority of compute will be front loaded into a startup cost as SERUM learns a user’s workflow, with minor revisions after the incubation period. This frees up compute for live suggestions. Evaluation in live assistive settings and scaling to larger video corpora are the highest-priority directions for future work.

Counterfactual scenarios. Future work could explore reversing SERUM to allow video generation models to imagine counterfactual scenarios. While SERUM infers actions and intentions from video, the reverse would use SERUM’s action and intent labels to generate video. This would enable generating counterfactual videos through perturbing inferred actions and intentions.

Hallucinations. Over SERUM’s iterative annotation passes, we observe two main sources of hallucinations: 1. The image is not clear (e.g., due to motion blur, occlusion). 2. The VLM confuses whether an action is starting or ending due to limited temporal granularity. For most hallucinations SERUM self-corrects by re-examining the original frame in each pass and by attaining neighbor consensus via the temporal context window to normalize inconsistent cases.

Acknowledgements.

We thank the members of the Minnesota NLP group for giving feedback on initial drafts and, crucially, our colleague-annotators who dedicated significant time annotating SERUM’s results over several trials.

Ethics Statement

This work analyzes publicly available YouTube videos and does not involve human subjects research. We acknowledge that behavioral modeling from screen recordings could be misused for unauthorized surveillance; our work is intended for user-initiated workflow analysis and support. We release our code to promote reproducibility and encourage its responsible use. Annotations are generated by a vision-language model and may reflect biases present in its training data.

Human annotation. Two forms of human annotation supported this work: (1) a calibration set of 200 label pairs (100 activities, 100 intents) was hand-rated by one of the authors to fit the SentenceBERT semantic-merge threshold t^* (§6a); (2) five members of our research lab rated final-pass labels for accuracy on a set of 26 videos pre-vetted by the authors (§4.4). Annotators were uncompensated lab volunteers, viewed only the pre-vetted videos, and agreed to participate and to the use of their judgements in this research. No personally identifying information was collected from annotators, and the videos contained no third-party private data. We did not seek formal Institutional Review Board approval, treating the rating task as internal validation by research collaborators; we acknowledge this is a limitation of the human evaluation and that a small, in-lab annotator pool may bias results toward positive judgments.

LLM Disclosure

In accordance with the COLM 2026 policy on LLM usage, we disclose the following. LLM-assisted coding tools were used during software development and infrastructure management. An LLM was also used to proofread drafts and assist with an initial literature survey; all references were verified by the authors.

LLMs were not used to generate experimental results, figures, datasets, or quantitative analysis. The research ideas, experimental design, implementation, analysis, and paper content are the work of the authors.

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A Full Data Collection

Table 7: Dataset Overview

Video	Category	Frames	Passes	Interval	Source URL
ACS_salestrainingP12	Daily Life	55	12	5s	youtu.be/ZG4ExqMVA7w
AC_leetcode2P12	Coding	80	12	5s	youtu.be/vRAK2YnFr1o
AC_leetcodeP12	Coding	352	12	5s	youtu.be/zeLZuhi6eYU
AC_pizzaP12	Cooking	86	12	5s	youtu.be/Q9j6HhF0tGE
AC_profreactsP12	Daily Life	108	12	5s	youtu.be/3mRvCF4qyTA
AC_sandwichP12	Cooking	120	12	5s	youtu.be/ad8TWumCSnY
AC_studrecordingP12	Daily Life	100	12	5s	youtu.be/eB54LIupAhU
AC_ukdayinlifeP12	Daily Life	187	12	5s	youtu.be/BvWnEi0oAEk
AC_waiterP12	Cooking	133	12	5s	youtu.be/w4pGt-iGpBI
BC_dunkinhelpP12	Cooking	213	12	5s	youtu.be/j_gUBLwxG1U
BC_nycswevlogP12	Daily Life	114	12	5s	youtu.be/4lo81zt7HK8
BC_pizzarushP12	Cooking	252	12	5s	youtu.be/S5ltPbUur38
BC_swevlogP12	Daily Life	121	12	5s	youtu.be/b_eeMSN097U
BC_vibecodingP12	Coding	157	12	5s	youtu.be/P3JA7MTiGg8
CC_baristaP12	Cooking	429	12	5s	youtu.be/jdguVU0F7fs
CC_swissweP12	Daily Life	327	12	5s	youtu.be/_GSI2RaiV0s
DC_calcappcodingP12	Coding	409	12	5s	youtu.be/sBJmRD7kNTk
DC_snakecodingP12	Coding	289	12	5s	youtu.be/Wlu4MsBnjuk
PERS_coinflipP12	Coding	137	12	5s	youtu.be/-o-H1Ecqo_M
PERS_movieP12	Coding	200	12	5s	youtu.be/J6uam9jEmDU
PERS_weatherP12	Coding	246	12	5s	youtu.be/iILFBGm_I9M
bartenderP12	Cooking	212	12	5s	youtu.be/1G-9Pibx5JI
basketballP12	Physical	50	12	5s	youtu.be/N7RNoIeA7Sk
behindP12	Daily Life	124	12	5s	youtu.be/h4exLX8Wz4E
carrepair2P12	Physical	113	12	5s	youtu.be/o00BJCfAf0Y
carrepair3P12	Physical	171	12	5s	youtu.be/VdR5zPyqp_4
carrepairP12	Physical	223	12	5s	youtu.be/vHdz74orr1Q
cashboothP12	Cooking	50	12	5s	youtu.be/9lNBUsF4WRU
coding2P12	Coding	185	12	5s	youtu.be/gRyvG7PZ4m0
coding3P12	Coding	226	12	5s	youtu.be/825u2Puaej0
codingP12	Coding	179	12	5s	youtu.be/DfDPJqD3FjI
codinglogoP12	Coding	234	12	5s	youtu.be/B_puD1rTs0Q
codingqrcoedP12	Coding	198	12	5s	youtu.be/I50Xwve6QW4
competitiveP12	Coding	206	12	5s	youtu.be/uGrBHohIgQY
compgamingP12	Coding	127	12	5s	youtu.be/yCezqhatLV8
construction2P12	Physical	81	12	5s	youtu.be/G1sCRChrdFU
constructionP12	Physical	260	12	5s	youtu.be/2avgoVsQ_og
csscodingP12	Coding	42	12	5s	youtu.be/EZHPsuIXawk
dayinthelifesweP12	Daily Life	102	12	5s	youtu.be/aTHBJwVgu3I
drivingP12	Physical	158	12	5s	youtu.be/iSnP5c997Uk
dunkinP12	Cooking	210	12	5s	youtu.be/hEJaSuDiQU8
fluttercodingP12	Coding	216	12	5s	youtu.be/C7Kafde7gZ4

Continued on next page

Table 7: Dataset Overview (continued)

Video	Category	Frames	Passes	Interval	Source URL
goprochefP12	Cooking	299	12	5s	youtu.be/CBSsL4u_nng
headchefP12	Cooking	350	12	5s	youtu.be/Ipe9xJCfuTM
hotdogP12	Cooking	265	12	5s	youtu.be/YMpGWAB411I
labworkP12	Daily Life	164	12	5s	youtu.be/C3aKnhXn20U
markiplierP12	Daily Life	317	12	5s	youtu.be/Yk-I7IVLAGo
mcdcookP12	Cooking	151	12	5s	youtu.be/8kcUsQdxtSs
mcdtakingordersP12	Cooking	337	12	5s	youtu.be/_c8PppBiMqE
microbialP12	Daily Life	54	12	5s	youtu.be/NUkrCXMd13o
musicplayercodingP12	Coding	283	12	5s	youtu.be/KndQpfPk00Y
paperworkP12	Daily Life	69	12	5s	youtu.be/JdkMmLhPw_E
phonerepairP12	Physical	343	12	5s	youtu.be/p9hA59nn7uQ
radiatorrepairP12	Physical	98	12	5s	youtu.be/ldIo1L6S_Sw
rmlineP12	Physical	105	12	5s	youtu.be/jIJTEm0qNuo
sushiP12	Cooking	183	12	5s	youtu.be/KUzYFMgWs4w
tractorfarmingP12	Physical	96	12	5s	youtu.be/rqA-iT2DK04
tttcodingP12	Coding	175	12	5s	youtu.be/MgtGHfdpigU
tutorialP12	Daily Life	92	12	5s	youtu.be/a32fbqPNir4
welshgardeningP12	Physical	160	12	5s	youtu.be/T3fgL091hXs
wsinstallP12	Daily Life	102	12	5s	youtu.be/QadguqFAt_8

B Verbose Next-Action Prediction Task Results

Table 8: Markov Prediction Accuracy (Final Pass)

Video	Type	Vocab	Markov	Majority	Wt. Rand	Uniform	Vocab ⁿ	Markov ⁿ	Maj ⁿ
ACS_salestrainingP12	Act	7	61.9%	19.0%	28.6%	14.3%	6	61.9%	19.0%
ACS_salestrainingP12	Int	24	4.8%	0.0%	4.3%	4.2%	7	9.5%	38.1%
AC_leetcode2P12	Act	2	100.0%	100.0%	96.0%	50.0%	2	100.0%	100.0%
AC_leetcode2P12	Int	8	83.9%	83.9%	40.0%	12.5%	5	100.0%	100.0%
AC_leetcodeP12	Act	6	92.6%	94.3%	84.5%	16.7%	5	93.4%	94.3%
AC_leetcodeP12	Int	11	63.1%	69.7%	41.0%	9.1%	6	65.6%	69.7%
AC_pizzaP12	Act	45	5.9%	0.0%	2.1%	2.2%	25	8.8%	0.0%
AC_pizzaP12	Int	43	0.0%	5.9%	2.8%	2.3%	13	52.9%	32.4%
AC_profreactsP12	Act	22	11.6%	20.9%	10.5%	4.5%	19	16.3%	20.9%
AC_profreactsP12	Int	40	7.0%	23.3%	4.4%	2.5%	26	14.0%	25.6%
AC_sandwichP12	Act	46	4.3%	0.0%	1.4%	2.2%	23	12.8%	12.8%
AC_sandwichP12	Int	19	51.1%	48.9%	30.3%	5.3%	7	74.5%	74.5%
AC_studrecordingP12	Act	20	48.7%	25.6%	8.6%	5.0%	14	51.3%	7.7%
AC_studrecordingP12	Int	19	59.0%	23.1%	10.4%	5.3%	8	64.1%	28.2%
AC_ukdayinlifeP12	Act	44	12.2%	5.4%	3.3%	2.3%	31	24.3%	5.4%
AC_ukdayinlifeP12	Int	66	5.4%	1.4%	1.9%	1.5%	37	17.6%	9.5%
AC_waiterP12	Act	24	20.8%	24.5%	10.2%	4.2%	15	32.1%	39.6%
AC_waiterP12	Int	18	28.3%	39.6%	20.4%	5.6%	7	56.6%	47.2%
BC_dunkinhelpP12	Act	66	17.6%	25.9%	6.4%	1.5%	29	50.6%	52.9%
BC_dunkinhelpP12	Int	42	15.3%	22.4%	9.5%	2.4%	13	65.9%	71.8%
BC_nycswevlogP12	Act	50	6.7%	6.7%	2.8%	2.0%	45	6.7%	6.7%
BC_nycswevlogP12	Int	59	6.7%	4.4%	1.6%	1.7%	47	6.7%	4.4%
BC_pizzarushP12	Act	67	31.0%	5.0%	5.0%	1.5%	28	52.0%	13.0%
BC_pizzarushP12	Int	38	53.0%	26.0%	15.1%	2.6%	15	86.0%	87.0%
BC_swevlogP12	Act	31	12.5%	20.8%	7.3%	3.2%	24	29.2%	22.9%
BC_swevlogP12	Int	43	6.2%	8.3%	4.1%	2.3%	21	8.3%	8.3%
BC_vibecodingP12	Act	9	43.5%	50.0%	31.5%	11.1%	8	62.9%	50.0%
BC_vibecodingP12	Int	18	40.3%	43.5%	31.2%	5.6%	8	48.4%	43.5%
CC_baristaP12	Act	94	18.1%	21.1%	6.2%	1.1%	37	32.7%	28.7%
CC_baristaP12	Int	83	7.6%	0.0%	2.5%	1.2%	20	48.5%	33.3%
CC_swissweP12	Act	89	5.4%	7.7%	2.2%	1.1%	53	8.5%	8.5%
CC_swissweP12	Int	105	20.0%	21.5%	2.6%	1.0%	64	24.6%	21.5%
DC_calcappcodingP12	Act	6	82.2%	82.8%	55.2%	16.7%	6	82.2%	82.8%
DC_calcappcodingP12	Int	86	1.8%	0.0%	0.5%	1.2%	19	90.8%	91.4%
DC_snakecodingP12	Act	5	10.4%	11.3%	11.5%	20.0%	3	11.3%	11.3%
DC_snakecodingP12	Int	9	11.3%	14.8%	13.0%	11.1%	2	99.1%	99.1%
PERS_coinflipP12	Act	10	63.0%	59.3%	45.0%	10.0%	6	72.2%	59.3%
PERS_coinflipP12	Int	33	57.4%	0.0%	4.6%	3.0%	10	92.6%	94.4%
PERS_movieP12	Act	6	100.0%	100.0%	88.9%	16.7%	5	100.0%	100.0%
PERS_movieP12	Int	36	43.0%	8.9%	8.5%	2.8%	9	72.2%	13.9%
PERS_weatherP12	Act	8	63.3%	62.2%	33.1%	12.5%	5	63.3%	62.2%
PERS_weatherP12	Int	28	34.7%	0.0%	8.4%	3.6%	4	95.9%	95.9%
bartenderP12	Act	85	10.7%	2.4%	1.9%	1.2%	45	11.9%	3.6%
bartenderP12	Int	96	10.7%	11.9%	1.8%	1.0%	40	28.6%	25.0%
basketballP12	Act	10	21.1%	21.1%	13.8%	10.0%	7	42.1%	52.6%
basketballP12	Int	9	89.5%	89.5%	41.8%	11.1%	5	100.0%	100.0%
behindP12	Act	15	12.2%	0.0%	3.9%	6.7%	14	12.2%	0.0%
behindP12	Int	19	8.2%	0.0%	2.2%	5.3%	13	8.2%	0.0%
carrepair2P12	Act	32	8.9%	13.3%	4.6%	3.1%	19	8.9%	13.3%
carrepair2P12	Int	31	31.1%	44.4%	12.8%	3.2%	14	53.3%	57.8%
carrepair3P12	Act	44	2.9%	5.9%	2.2%	2.3%	20	20.6%	22.1%
carrepair3P12	Int	38	13.2%	11.8%	8.5%	2.6%	16	29.4%	38.2%
carrepairP12	Act	52	1.1%	0.0%	1.4%	1.9%	34	2.2%	0.0%
carrepairP12	Int	33	32.6%	30.3%	12.7%	3.0%	15	46.1%	47.2%
cashboothP12	Act	10	10.5%	21.1%	10.5%	10.0%	8	15.8%	26.3%
cashboothP12	Int	14	31.6%	10.5%	12.0%	7.1%	6	42.1%	47.4%
coding2P12	Act	14	8.2%	0.0%	6.2%	7.1%	9	16.4%	0.0%

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Table 8 – continued

Video	Type	Vocab	Markov	Majority	Wt. Rand	Uniform	Vocab ⁿ	Markov ⁿ	Maj ⁿ
coding2P12	Int	62	8.2%	0.0%	2.7%	1.6%	16	16.4%	0.0%
coding3P12	Act	10	68.9%	66.7%	54.3%	10.0%	7	66.7%	66.7%
coding3P12	Int	8	90.0%	90.0%	77.1%	12.5%	5	100.0%	100.0%
codingP12	Act	4	100.0%	100.0%	91.0%	25.0%	3	100.0%	100.0%
codingP12	Int	17	42.3%	16.9%	16.5%	5.9%	4	90.1%	90.1%
codinglogoP12	Act	2	98.9%	98.9%	98.2%	50.0%	2	98.9%	98.9%
codinglogoP12	Int	2	100.0%	100.0%	98.6%	50.0%	2	100.0%	100.0%
codingqrcodeP12	Act	8	89.9%	93.7%	76.6%	12.5%	6	89.9%	93.7%
codingqrcodeP12	Int	42	32.9%	48.1%	10.5%	2.4%	9	60.8%	75.9%
competitiveP12	Act	5	72.0%	76.8%	72.8%	20.0%	4	73.2%	76.8%
competitiveP12	Int	12	37.8%	26.8%	21.3%	8.3%	3	68.3%	74.4%
compgamingP12	Act	3	96.0%	96.0%	84.1%	33.3%	3	96.0%	96.0%
compgamingP12	Int	2	76.0%	76.0%	62.0%	50.0%	1	–	–
construction2P12	Act	24	21.9%	28.1%	10.5%	4.2%	12	21.9%	31.2%
construction2P12	Int	26	12.5%	34.4%	9.2%	3.8%	5	71.9%	71.9%
constructionP12	Act	13	71.8%	75.7%	49.9%	7.7%	9	72.8%	75.7%
constructionP12	Int	18	22.0%	26.8%	21.9%	5.6%	7	81.7%	81.7%
csscodingP12	Act	3	87.5%	87.5%	50.4%	33.3%	2	93.8%	87.5%
csscodingP12	Int	8	37.5%	0.0%	19.1%	12.5%	4	62.5%	62.5%
dayinthelifesweP12	Act	42	0.0%	7.5%	2.2%	2.4%	31	5.0%	12.5%
dayinthelifesweP12	Int	48	12.5%	15.0%	2.3%	2.1%	31	12.5%	0.0%
drivingP12	Act	21	81.0%	84.1%	25.5%	4.8%	14	85.7%	92.1%
drivingP12	Int	17	93.7%	95.2%	46.7%	5.9%	11	95.2%	95.2%
dunkinP12	Act	98	1.2%	2.4%	1.3%	1.0%	39	10.8%	13.3%
dunkinP12	Int	74	6.0%	14.5%	2.9%	1.4%	25	10.8%	22.9%
fluttercodingP12	Act	14	62.8%	68.6%	43.8%	7.1%	9	65.1%	68.6%
fluttercodingP12	Int	55	30.2%	37.2%	5.9%	1.8%	22	45.3%	54.7%
goprochefP12	Act	112	3.4%	10.1%	2.3%	0.9%	53	21.8%	31.9%
goprochefP12	Int	68	14.3%	21.8%	6.5%	1.5%	25	23.5%	27.7%
headchefP12	Act	81	15.1%	18.0%	4.4%	1.2%	35	23.0%	19.4%
headchefP12	Int	59	34.5%	42.4%	15.1%	1.7%	22	39.6%	42.4%
hotdogP12	Act	70	18.1%	24.8%	5.4%	1.4%	30	54.3%	48.6%
hotdogP12	Int	48	12.4%	19.0%	6.9%	2.1%	15	82.9%	83.8%
labworkP12	Act	8	67.7%	67.7%	44.8%	12.5%	6	70.8%	70.8%
labworkP12	Int	9	46.2%	38.5%	27.1%	11.1%	2	47.7%	47.7%
markiplierP12	Act	22	46.0%	49.2%	30.1%	4.5%	15	66.7%	67.5%
markiplierP12	Int	53	19.0%	23.8%	9.2%	1.9%	26	39.7%	46.0%
mcdcookP12	Act	30	25.0%	26.7%	8.9%	3.3%	9	80.0%	80.0%
mcdcookP12	Int	21	28.3%	38.3%	14.1%	4.8%	4	95.0%	95.0%
mcddtakingordersP12	Act	63	20.9%	16.4%	5.1%	1.6%	42	29.9%	24.6%
mcddtakingordersP12	Int	66	11.9%	1.5%	3.5%	1.5%	35	21.6%	4.5%
microbialP12	Act	17	4.8%	0.0%	8.1%	5.9%	9	19.0%	19.0%
microbialP12	Int	6	52.4%	61.9%	45.6%	16.7%	2	90.5%	95.2%
musicplayercodingP12	Act	8	92.9%	94.7%	83.5%	12.5%	5	92.9%	94.7%
musicplayercodingP12	Int	43	11.5%	10.6%	4.3%	2.3%	9	61.9%	12.4%
paperworkP12	Act	15	11.1%	11.1%	8.8%	6.7%	8	33.3%	29.6%
paperworkP12	Int	7	22.2%	33.3%	27.8%	14.3%	3	44.4%	44.4%
phonerepairP12	Act	61	15.3%	17.5%	5.8%	1.6%	30	20.4%	19.0%
phonerepairP12	Int	31	34.3%	10.9%	9.6%	3.2%	15	56.9%	64.2%
radiatorrepairP12	Act	9	53.8%	64.1%	30.7%	11.1%	5	76.9%	84.6%
radiatorrepairP12	Int	5	41.0%	46.2%	36.8%	20.0%	3	100.0%	100.0%
rmlineP12	Act	10	7.3%	2.4%	11.6%	10.0%	5	48.8%	51.2%
rmlineP12	Int	10	65.9%	68.3%	34.3%	10.0%	3	82.9%	82.9%
sushiP12	Act	92	1.4%	6.8%	1.2%	1.1%	26	39.7%	50.7%
sushiP12	Int	38	21.9%	20.5%	8.3%	2.6%	11	78.1%	79.5%
tractorfarmingP12	Act	25	7.9%	0.0%	4.0%	4.0%	18	78.9%	84.2%
tractorfarmingP12	Int	34	2.6%	0.0%	2.9%	2.9%	19	81.6%	0.0%
tttcodingP12	Act	6	72.5%	72.5%	54.9%	16.7%	6	72.5%	72.5%
tttcodingP12	Int	13	100.0%	100.0%	75.4%	7.7%	4	100.0%	100.0%
tutorialP12	Act	13	47.2%	52.8%	15.8%	7.7%	10	72.2%	58.3%
tutorialP12	Int	31	50.0%	41.7%	8.6%	3.2%	16	63.9%	16.7%

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Table 8 – continued

Video	Type	Vocab	Markov	Majority	Wt. Rand	Uniform	Vocab ⁿ	Markov ⁿ	Maj ⁿ
welshgardeningP12	Act	42	25.4%	38.1%	7.7%	2.4%	20	60.3%	68.3%
welshgardeningP12	Int	39	44.4%	19.0%	8.7%	2.6%	14	58.7%	28.6%
wslinstallP12	Act	27	22.5%	42.5%	10.3%	3.7%	21	25.0%	47.5%
wslinstallP12	Int	49	2.5%	5.0%	1.5%	2.0%	24	2.5%	0.0%

ⁿ = normalized labels

C Generalizing Procedure on EPIC-KITCHENS-100

Table 9: EPIC-KITCHENS-100 generalization (366 videos from 37 participants). Same Markov harness as Table 2; final-pass P11 (activity) / P12 (intent). ⁿ denotes models built on normalized labels.

Model	Activity		Intent	
	Top-1 ($\uparrow$)	PPL ($\downarrow$)	Top-1 ($\uparrow$)	PPL ($\downarrow$)
Markov	15.3 $\pm$ 18.9	27.4 $\pm$ 13.5	33.1 $\pm$ 27.6	14.4 $\pm$ 9.1
Majority	14.5 $\pm$ 20.1	42.2 $\pm$ 25.3	31.6 $\pm$ 29.5	20.4 $\pm$ 16.6
Wt. Random	6.4 $\pm$ 7.6	42.2 $\pm$ 25.3	17.6 $\pm$ 17.7	20.4 $\pm$ 16.6
Uniform	4.1 $\pm$ 2.6	31.2 $\pm$ 12.7	7.4 $\pm$ 6.3	19.1 $\pm$ 8.9
Markov ⁿ	28.3 $\pm$ 25.0	13.8 $\pm$ 7.6	53.6 $\pm$ 28.7	5.5 $\pm$ 3.5
Majority ⁿ	22.4 $\pm$ 25.4	21.5 $\pm$ 15.1	47.0 $\pm$ 32.9	7.7 $\pm$ 6.9

366 videos, 33,788 frames at the final activity pass.

Table 10: Curated vs. EPIC-KITCHENS-100 generalization (61 curated videos vs. 366 EK videos; same Markov harness). Absolute Markovⁿ accuracy is lower on EK; the Markovⁿ–Majorityⁿ method gap is wider on EK. Δ = EK – Curated. ⁿ denotes models built on normalized labels.

Metric	Curated	EK	Δ
<i>Absolute Markovⁿ performance (curated wins on accuracy; EK wins on intent PPL):</i>			
Activity Top-1 (%)	47.9 $\pm$ 29.2	28.3 $\pm$ 25.0	-19.5 pp
Activity PPL	10.7 $\pm$ 10.8	13.8 $\pm$ 7.6	+3.1
Intent Top-1 (%)	60.7 $\pm$ 30.8	53.6 $\pm$ 28.7	-7.1 pp
Intent PPL	7.4 $\pm$ 9.4	5.5 $\pm$ 3.5	-1.9
<i>Markovⁿ–Majorityⁿ method gap (EK gap is wider on both):</i>			
Activity (pp)	+4.9	+5.9	+1.0 pp
Intent (pp)	+1.5	+6.5	+5.1 pp

Table 11: Per-participant Markovⁿ top-1 accuracy and perplexity on EPIC-KITCHENS-100 (final-pass P11/P12, normalized labels).

Participant	Videos	Act Top-1	Act PPL	Int Top-1	Int PPL
P04	28	20.9 $\pm$ 23.5	15.4 $\pm$ 8.1	42.3 $\pm$ 26.9	7.2 $\pm$ 3.7
P22	27	23.3 $\pm$ 14.8	16.4 $\pm$ 7.1	45.3 $\pm$ 23.9	6.6 $\pm$ 4.0
P02	23	28.6 $\pm$ 30.8	15.7 $\pm$ 7.8	59.3 $\pm$ 30.2	5.3 $\pm$ 3.5
P03	23	27.6 $\pm$ 22.2	11.2 $\pm$ 5.2	58.2 $\pm$ 26.8	4.7 $\pm$ 3.0
P08	17	14.8 $\pm$ 11.6	18.6 $\pm$ 8.7	51.0 $\pm$ 28.2	6.5 $\pm$ 3.5
P28	17	32.8 $\pm$ 18.3	11.9 $\pm$ 6.6	53.1 $\pm$ 34.8	5.1 $\pm$ 3.6
P30	17	15.8 $\pm$ 16.5	18.2 $\pm$ 8.0	36.5 $\pm$ 29.6	7.6 $\pm$ 3.9
P01	16	29.9 $\pm$ 26.7	14.5 $\pm$ 7.3	58.9 $\pm$ 27.3	5.0 $\pm$ 3.0
P07	16	22.3 $\pm$ 15.9	12.0 $\pm$ 7.0	56.6 $\pm$ 30.8	4.8 $\pm$ 3.6
P26	16	42.2 $\pm$ 27.3	7.0 $\pm$ 3.8	70.9 $\pm$ 28.6	2.7 $\pm$ 1.7
P06	13	34.0 $\pm$ 29.7	13.2 $\pm$ 6.5	48.5 $\pm$ 30.9	5.7 $\pm$ 3.2
P25	12	28.2 $\pm$ 22.7	12.0 $\pm$ 5.6	51.2 $\pm$ 22.2	5.4 $\pm$ 2.8
P11	10	19.8 $\pm$ 15.8	15.8 $\pm$ 4.1	47.0 $\pm$ 24.8	6.2 $\pm$ 3.1
P12	10	29.7 $\pm$ 12.8	13.3 $\pm$ 4.7	38.7 $\pm$ 24.9	6.8 $\pm$ 3.5
P27	10	32.0 $\pm$ 27.6	13.5 $\pm$ 9.5	51.1 $\pm$ 29.1	6.2 $\pm$ 4.4
P31	9	23.4 $\pm$ 29.7	17.2 $\pm$ 9.4	64.0 $\pm$ 24.4	4.6 $\pm$ 2.2
P33	9	25.4 $\pm$ 22.8	14.9 $\pm$ 5.4	41.3 $\pm$ 26.7	6.4 $\pm$ 2.3
P35	9	33.6 $\pm$ 23.6	14.0 $\pm$ 8.4	56.3 $\pm$ 21.5	4.7 $\pm$ 2.2
P15	8	24.1 $\pm$ 14.5	15.1 $\pm$ 6.9	56.8 $\pm$ 25.7	4.9 $\pm$ 2.9
P09	7	28.3 $\pm$ 26.1	12.0 $\pm$ 6.7	46.2 $\pm$ 29.5	5.7 $\pm$ 3.4
P23	7	43.7 $\pm$ 37.8	10.7 $\pm$ 7.3	56.2 $\pm$ 30.7	5.4 $\pm$ 3.6

Participant	Videos	Act Top-1	Act PPL	Int Top-1	Int PPL
P24	7	36.4±17.1	12.0±6.2	62.3±26.9	3.9±2.1
P34	7	42.8±26.2	8.8±4.4	61.3±26.7	4.2±2.4
P05	6	47.4±29.0	9.4±5.5	79.0±16.0	2.8±1.3
P18	6	26.1±26.8	14.7±7.7	47.0±15.7	6.5±3.7
P20	5	18.2±17.4	17.8±6.4	65.4±23.5	4.1±1.8
P13	4	64.7±29.0	7.8±4.7	66.9±32.8	4.8±4.4
P29	4	26.1±34.1	13.2±8.8	71.8±26.5	3.6±3.0
P32	4	31.1±32.4	9.4±5.0	70.1±18.1	2.8±1.5
P10	3	23.4±3.5	17.2±1.4	38.3±13.6	8.6±3.3
P17	3	16.1±8.7	18.4±3.2	46.6±9.4	6.7±0.6
P19	3	22.8±23.2	17.9±8.7	41.0±35.0	5.5±3.1
P14	2	22.6±22.6	9.2±2.1	71.4±3.6	3.1±0.3
P16	2	34.0±6.4	11.4±2.5	71.3±20.2	3.1±1.1
P21	2	23.5±15.0	21.1±9.0	57.8±8.9	6.9±2.9
P36	2	87.2±2.1	2.1±0.1	89.4±0.0	1.6±0.2
P37	2	72.3±14.9	10.3±8.6	50.0±13.8	5.4±2.6
Overall	366	28.3±25.0	13.8±7.6	53.6±28.7	5.5±3.5

D Additional Qualitative Example

Content creation (tutorialP12). Figure 8 shows refinement on a content-creation video. Activity labels evolve from perceptual (*sitting, browsing web*) to task-specific (*preparing tutorial video on intersection observer API, responding to viewer comment*) by pass 11.

The intent graphs (d–f) reveal structure invisible in the activity graph. By pass 12, two workflow clusters emerge: an audience-facing loop (*responding to viewer comment* → *speaking into microphone* → *creating digital content* → *preparing tutorial video on intersection observer API*), and a production pipeline (*managing content schedule* → *reviewing and refining video content* → *managing video content pipeline*). These clusters connect through *managing content creation workflow*. An agent consuming this model could distinguish recording from planning phases — a distinction the activity graph cannot surface.

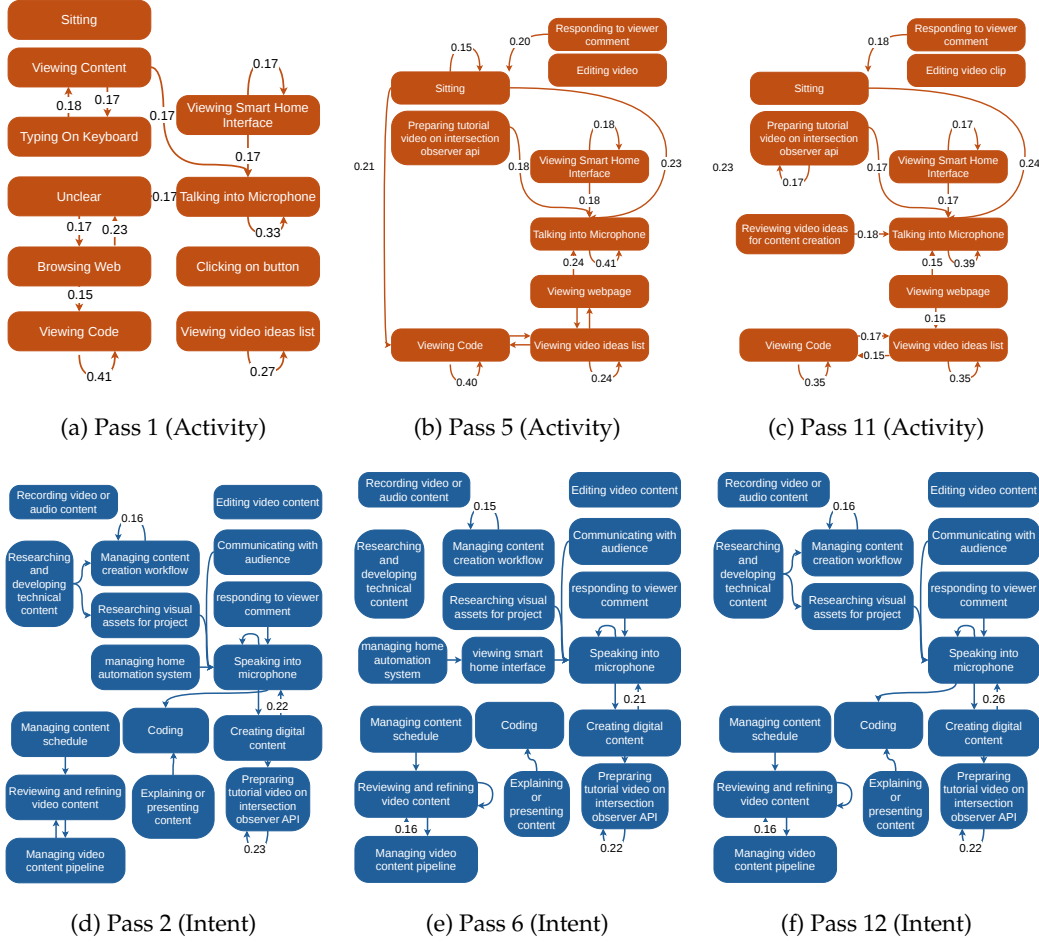


Figure 8: Content creation video (tutorialP12). *Top*: activity labels refine from perceptual to task-specific. *Bottom*: intent graphs reveal two workflow clusters (audience-facing vs. production) connected through a management hub.

E Pass-by-Pass Model Performance

Figure 9 shows accuracy and perplexity across all 12 passes (61 videos). Activity accuracy (a): raw Markov and Majority are closely matched; normalized Markov consistently leads. Intent accuracy (b): normalized models show clearer separation, reaching $\sim 60\%$ vs. $\sim 30\%$ raw accuracy. Perplexity (c, d): Markov achieves the lowest at every pass; normalization roughly halves it. Wide standard deviation bands reflect high per-video variance from vocabulary size and domain differences.

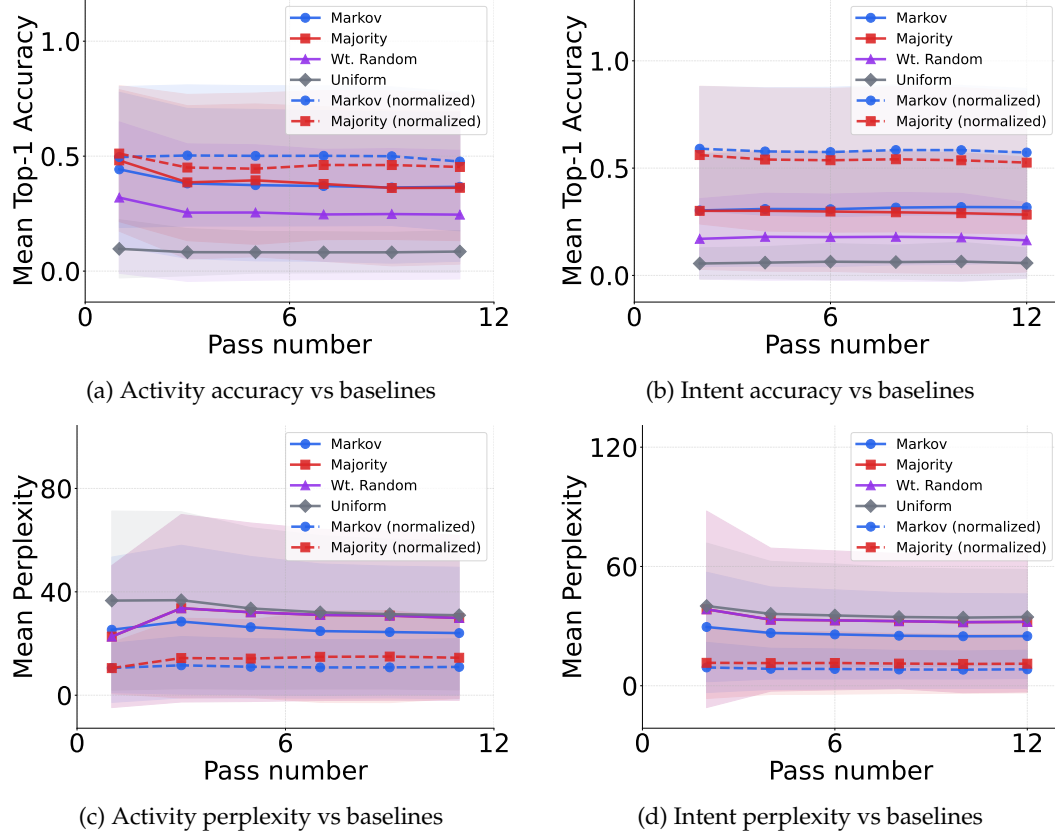


Figure 9: Pass-by-pass top-1 accuracy and perplexity for normalized Markov vs. baselines across all 12 annotation passes, averaged over 61 videos with standard deviation bands.

F Threshold Calibration Procedure

To calibrate the semantic merging threshold, we assemble all unique activity and intent labels across every video and pass, compute pairwise SentenceBERT cosine distances, and partition the distance range into 10 equal-width bins. We randomly sample 10 pairs per bin per type (activity, intent), yielding 100 pairs per type (200 total) stratified across the full similarity spectrum. A single annotator labels each pair as *same* (semantically equivalent), *different* (distinct states), or *skip* (ambiguous). The optimal threshold t^* is selected as the cosine distance maximizing F1 score on non-skipped pairs, treating *same* as the positive class.

G Prompt Templates

Table 12 summarizes the four prompt templates used across annotation passes. Pass 1 receives only the frame image; all subsequent passes additionally receive the temporal context window (RLE of neighboring frames’ labels) and the inter-pass summary.

Table 12: Prompt templates by pass type. Each prompt instructs the VLM to output structured JSON with a state label, confidence score (1–10), and supporting evidence.

Prompt	Passes	Key instruction
Activity (first)	P1	Identify the dominant observable action from the frame. Describe <i>what</i> is happening (e.g., <code>typing_on_keyboard</code>), not <i>why</i> . Use lowercase with underscores.
Intent (first)	P2	Given prior-pass activity labels and temporal context, infer the user’s underlying goal (e.g., <code>grocery_shopping</code>). Look for patterns across sequential activities. Assign lower confidence to speculative intents.
Refined activity	P3, P5, ...	Re-analyze with enriched context from prior activity <i>and</i> intent passes. Be more precise (e.g., <code>generic typing</code> + intent coding $\rightarrow$ <code>typing_code</code>). Collapse synonyms. Split actions that serve different intents. Report what changed and why.
Refined intent	P4, P6, ...	Re-analyze intents with multiple passes of context. Validate or invalidate prior inferences based on subsequent observations. Discover higher-level goal patterns. Report what changed and why.

Full prompt text is available in the released codebase ⁸

⁸<https://github.com/minnesotanlp/SERUM/>

H Model choice preliminary study

We investigated 3 new models over 4 videos chosen randomly from the expanded annotation round. We find each model still reaches schematic equilibrium at every scale tested, but generally larger models took longer to reach schematic equilibrium (Table 13). There is no obvious pattern to the effectiveness of larger models in next-state prediction. Larger models benefit more from normalization (32B’s Markov accuracy saw a 222% increase going from 2.7 to 6.0) (Table 14) primarily due to larger models being more verbose and specific about label assessments, thereby inflating vocabulary sizes (Table 13).

Table 13: Schematic equilibrium across VLM scales: vocab size by pass. 4 Qwen3-VL variants on 4 videos.

Video	Model	Intent passes						Activity passes					
		P2	P4	P6	P8	P10	P12	P1	P3	P5	P7	P9	P11
AC_leetcode	4B	18	18	18	18	18	17	5	6	6	6	6	6
	8B	14	14	14	14	14	14	6	6	6	6	6	6
	30B-A3B	5	5	5	5	5	5	8	7	7	7	7	7
	32B	18	15	15	13	14	14	11	15	10	12	12	12
AC_pizza	4B	42	33	30	30	30	31	67	70	68	67	67	67
	8B	55	46	43	42	42	38	54	54	52	42	48	44
	30B-A3B	41	40	41	40	40	40	60	60	59	59	59	59
	32B	67	58	51	44	39	39	73	63	56	52	47	44
AC_ukdayinlife	4B	82	69	67	65	65	64	84	92	90	87	85	83
	8B	93	78	68	64	65	61	59	58	53	51	49	51
	30B-A3B	72	73	70	70	68	68	73	80	80	79	78	77
	32B	119	88	70	59	52	48	98	95	81	73	68	63
BC_nycswelog	4B	71	66	63	63	63	63	63	65	66	65	66	67
	8B	85	69	66	62	56	56	44	53	53	46	50	46
	30B-A3B	74	69	69	68	69	69	67	72	72	72	72	72
	32B	101	77	66	62	60	60	77	73	62	66	65	67

Table 14: Model choice: Markov vs. majority accuracy (%) and perplexity before/after normalization, by label type. Δ = Markov – Majority. n = 4 videos, 4 Qwen3-VL variants.

Model	Raw				Normalized			
	Markov	Maj.	Δ	PPL	Markov	Maj.	Δ	PPL
<i>Activity (P11)</i>								
4B	25.7	27.4	-1.8	47.6	25.8	30.2	-4.4	28.3
8B	32.1	30.3	+1.8	30.4	33.5	29.5	+4.0	20.5
30B-A3B	17.4	12.3	+5.2	49.3	21.0	12.8	+8.2	28.8
32B	2.7	4.5	-1.8	44.0	6.0	6.8	-0.7	28.2
<i>Intent (P12)</i>								
4B	10.0	6.6	+3.4	39.8	17.4	8.8	+8.6	24.6
8B	23.7	21.4	+2.2	36.9	26.5	26.9	-0.4	22.4
30B-A3B	28.8	33.0	-4.3	37.7	38.8	41.0	-2.2	18.7
32B	5.7	8.7	-3.0	37.8	10.1	8.2	+1.9	23.5

I Transferability Study

As a preliminary check on cross-video transferability, we ran 3 leave-one-out markov evaluations on two same domain video triples: car repair, and coffee shop operation each (6 total). For car repair videos, Markov improved over the Majority baseline on the held-out video, demonstrating the transferability of learned models to new videos. On the other hand, in the coffee shop triple, one state (pouring_milk_into_cup) occurs very frequently (27%-39% frames in each video); in this case, Majority demonstrates better transferability.

Table 15: Cross-video Markov transferability (LOOCV): train on 2 videos’ concatenated activity sequences, test on the held-out video. Normalized vocabulary on the union of each triple at $t^* = 0.4268$.

Triple	Held-out test	Novel %	Markov	Majority	Markov – Majority
<i>Car repair</i>					
	carrepair	53.8%	15.3%	1.8%	+13.5
	carrepair2	53.1%	22.3%	8.0%	+14.3
	carrepair3	38.6%	14.1%	0.0%	+14.1
	<i>Mean</i>	—	—	—	+14.0
<i>Coffee shop</i>					
	dunkin	24.8%	21.1%	27.3%	-6.2
	BC_dunkinhelp	23.0%	25.9%	38.7%	-12.7
	CC_barista	38.7%	22.2%	28.5%	-6.3
	<i>Mean</i>	—	—	—	-8.4

We were also interested if there was significant pair-wise video transferability and conducted a brief pairwise transferability experiment, and found that yes if two videos are similar enough they are transferable; label ontology is relatively consistent for similar videos.

Table 16: Cross-video Markov transferability (pairwise): each video used as a single training set against another single video as test. More conservative than LOOCV (less train data, higher novel %).

Train	Test	Novel %	Markov	Majority	Markov – Majority
<i>Car repair</i>					
carrepair	→ carrepair2	53.1%	19.6%	1.8%	+17.9
carrepair	→ carrepair3	38.6%	7.6%	5.3%	+2.4
carrepair2	→ carrepair	69.5%	10.4%	0.0%	+10.4
carrepair2	→ carrepair3	66.7%	8.2%	0.0%	+8.2
carrepair3	→ carrepair	60.1%	10.4%	1.8%	+8.6
carrepair3	→ carrepair2	69.0%	13.4%	8.0%	+5.4
<i>Mean</i>		—	—	—	+8.8
<i>Coffee shop</i>					
dunkin	→ BC_dunkinhelp	31.5%	27.4%	38.7%	-11.3
dunkin	→ CC_barista	45.0%	21.3%	28.5%	-7.2
BC_dunkinhelp	→ dunkin	35.2%	21.1%	27.3%	-6.2
BC_dunkinhelp	→ CC_barista	47.8%	21.0%	28.5%	-7.5
CC_barista	→ dunkin	36.7%	19.1%	27.3%	-8.1
CC_barista	→ BC_dunkinhelp	32.9%	25.5%	38.7%	-13.2
<i>Mean</i>		—	—	—	-8.9